

ApproxLP: Approximate Credal Network Inference via Iterative Linearization

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Abstract

ApproxLP is an approximate inference algorithm for credal networks that reformulates the marginal inference problem as a multilinear program and solves it via iterative linearization (coordinate descent). At each step, all local conditional distributions except one are fixed, reducing the optimization to selecting the best extreme point for the remaining variable. This produces *inner* bounds—the returned interval is guaranteed to be contained within the true probability interval. ApproxLP is dramatically faster than exact variable elimination while often finding the exact bounds on small to medium networks.

1 Preliminaries

1.1 Multilinear Programming Formulation

The marginal inference task in a credal network asks for:

$$\underline{P}(Z=z \mid \mathbf{Y}=\mathbf{y}) = \min_{P \in \text{ext } K(\mathbf{x})} \frac{\sum_{\mathbf{x}'} \prod_i P(x_i \mid \pi_i)}{\sum_{z', \mathbf{x}'} \prod_i P(x_i \mid \pi_i)},$$

where the optimization is over all combinations of extreme points of the local credal sets. Using a tree decomposition and symbolic variable elimination (see `credal_ve.tex`), this can be reformulated as a multilinear program:

$$\begin{aligned} & \text{minimize} && f'_m(z), \\ & \text{subject to} && f'_m(Z) = t \cdot f_m(Z), \quad \sum_{z'} f'_m(z') = 1, \\ & && f_i = \sum_{x_i} \prod \{f \in \text{bucket}_i\}, \quad i = 1, \dots, m-1, \\ & && P(X_i \mid \pi_i) \in K(X_i \mid \pi_i), \quad \forall X_i, \pi_i. \end{aligned} \tag{1}$$

The multilinearity arises because the bucket constraints involve products of the optimization variables $P(X_i \mid \pi_i)$ and the intermediate functions f_i .

1.2 The Linearization Idea

Solving the multilinear program (1) directly is expensive. The key insight of Antonucci et al. (2013, 2015) is to apply **coordinate descent**: fix all local distributions $P(X_j \mid \pi_j)$ for $j \neq i$ to specific values, then the program becomes **linear** in the remaining variable $P(X_i \mid \pi_i)$.

Since the optimal solution of a linear program over a polytope is attained at a vertex, and we already have the extreme points of each credal set $K(X_i \mid \pi_i)$ from LRS vertex enumeration, the linear optimization step reduces to:

Try each extreme point of $K(X_i | \pi_i)$ and pick the one that optimizes the objective.

No LP solver is needed—vertex enumeration suffices.

2 Algorithm

Algorithm 1 ApproxLP: Coordinate Descent over Extreme Points (All Non-Evidence Variables)

Require: Credal factors with extreme points, evidence $\mathbf{Y}=\mathbf{y}$, max iterations T

Ensure: Inner bounds $[\underline{P}(Z=z | \mathbf{y}), \bar{P}(Z=z | \mathbf{y})]$ for **every** non-evidence variable Z and state $z \in D_Z$

```

1: for each variable  $Z$  in the network do
2:   if  $Z \in \mathbf{Y}$  then
3:      $\underline{P}(Z=z | \mathbf{y}), \bar{P}(Z=z | \mathbf{y}) \leftarrow \mathbb{K}[z = y_Z]$  for each  $z \in D_Z$  ▷ Evidence: point mass
4:   continue
5:   end if
6:    $\underline{\mathbf{p}} \leftarrow \mathbf{1}, \quad \bar{\mathbf{p}} \leftarrow \mathbf{0}$  ▷ Componentwise bounds over  $D_Z$ 
7:   for  $\text{sense} \in \{\min, \max\}$  do ▷ INITIALIZATION

8:     for each variable  $X_i$ , each parent config  $\pi$  do
9:        $P(X_i | \pi) \leftarrow \text{center}(\text{ext } K(X_i | \pi))$  ▷ Average of extreme points
10:    end for
11:     $p^* \leftarrow \text{VE}(\{P(X_i | \pi)\}, Z, \mathbf{y})$  ▷ Precise VE with fixed distributions
12:    for  $t = 1, \dots, T$  do ▷ COORDINATE DESCENT
13:       $\text{improved} \leftarrow \text{false}$ 
14:      for each variable  $X_i$ , each parent config  $\pi$  of  $X_i$  do
15:         $P_{\text{best}} \leftarrow P(X_i | \pi)$  ▷ Current incumbent
16:        for each vertex  $v \in \text{ext } K(X_i | \pi)$  do
17:           $P(X_i | \pi) \leftarrow v$ 
18:           $p_v \leftarrow \text{VE}(\{P(X_j | \pi_j)\}, Z, \mathbf{y})$ 
19:          if  $[\text{sense} = \min \wedge p_v(z) < p^*(z)] \vee [\text{sense} = \max \wedge p_v(z) > p^*(z)]$  then ▷  $z = \text{target}$ 
            state tracked by this run
20:             $p^* \leftarrow p_v, \quad P_{\text{best}} \leftarrow v, \quad \text{improved} \leftarrow \text{true}$ 
21:          end if
22:        end for
23:         $P(X_i | \pi) \leftarrow P_{\text{best}}$ 
24:      end for
25:      if  $\neg \text{improved}$  then break ▷ Converged
26:    end if
27:  end for
28:   $\underline{\mathbf{p}} \leftarrow \min(\underline{\mathbf{p}}, p^*), \quad \bar{\mathbf{p}} \leftarrow \max(\bar{\mathbf{p}}, p^*)$  ▷ Componentwise across  $D_Z$ 
29:  end for
30:  Record  $[\underline{P}(Z=z | \mathbf{y}), \bar{P}(Z=z | \mathbf{y})] \leftarrow [\underline{p}(z), \bar{p}(z)]$  for each  $z \in D_Z$ 
31: end for
32: return all recorded bounds

```

2.1 Computing Both Bounds

For each non-evidence variable Z , Algorithm 1 runs coordinate descent twice:

- With sense = min to drive $p^*(z)$ toward $\underline{P}(Z=z \mid \mathbf{y})$.
- With sense = max to drive $p^*(z)$ toward $\bar{P}(Z=z \mid \mathbf{y})$.

Each run produces a full probability vector $p^* \in \Delta^{|D_Z|}$; the final per-state bounds are taken componentwise across the two runs, which lets a single pair of runs yield intervals for every state $z \in D_Z$ simultaneously.

2.2 Standard VE with Fixed Distributions

The subroutine $\text{VE}(\{P(X_i \mid \pi)\}, Z, \mathbf{y})$ runs classical variable elimination on the precise Bayesian network defined by the fixed distributions. This is a standard operation:

1. Build a factor for each variable: a numpy array over $\{X_i\} \cup \text{Pa}(X_i)$ filled from $P(X_i \mid \pi)$.
2. Add indicator factors for evidence variables.
3. Eliminate all variables except Z by combining and marginalizing.
4. Normalize and return $P(Z \mid \mathbf{y})$.

Since all distributions are fixed (no sets of functions), each VE run is very fast.

3 Compact Description (NeurIPS Format)

Overview. ApproxLP computes marginal probability bounds in a credal network $\langle G, \mathcal{K} \rangle$ — a DAG whose local conditional distributions $P(X_i \mid \pi_i)$ are each constrained to lie in a finitely-generated credal set $K(X_i \mid \pi_i)$ with pre-enumerated vertex set $\text{ext } K(X_i \mid \pi_i)$. For a given non-evidence variable Z , the lower bound $\underline{P}(Z=z \mid \mathbf{y})$ is the minimum of the posterior $P(Z=z \mid \mathbf{y})$ over all combinations of extreme points, one per (X_i, π_i) pair. Holding every local distribution fixed except one turns this global optimization into a linear program over a polytope whose vertices are already known, and a standard-LP result places the optimum at one of those vertices. ApproxLP exploits this: it fixes all distributions, sweeps through each (X_i, π_i) in turn, and replaces $P(X_i \mid \pi_i)$ with whichever of its extreme points most improves the objective — evaluated by running precise variable elimination on the fully-determined Bayesian network. No LP solver is needed.

What Algorithm 1 does. For every non-evidence variable Z , the algorithm runs coordinate descent twice — once with sense min and once with sense max. Each run starts by initializing all $P(X_i \mid \pi_i)$ at the centroid of their respective credal sets and computing $p^* = \text{VE}(\{P\}, Z, \mathbf{y})$, the full posterior vector on D_Z under these fixed distributions. A *sweep* then visits every (X_i, π_i) : for each vertex $v \in \text{ext } K(X_i \mid \pi_i)$, it temporarily sets $P(X_i \mid \pi_i) \leftarrow v$, re-evaluates VE, and accepts v (strictly, else rejects) if the target component $p^*(z)$ moves in the sense’s direction. Sweeps repeat until one produces no improvement or the cap T is reached. Because each VE call returns a full vector, the two runs together yield componentwise bounds $[\underline{p}(z), \bar{p}(z)]$ for every state $z \in D_Z$ simultaneously. Evidence variables are skipped by a point-mass shortcut. The output is *inner* — coordinate descent explores only a subset of feasible vertex combinations, so the returned interval is contained in the true interval.

Problem. Let $\langle G, \mathcal{K} \rangle$ be a credal network: a DAG G over variables $X_1, \dots, X_n$ whose conditional distributions are each known only to belong to a finitely-generated credal set $K(X_i | \pi_i)$ with vertex set $\text{ext } K(X_i | \pi_i)$, one per parent configuration π_i . Given evidence $\mathbf{Y} = \mathbf{y}$, we seek tight bounds $[\underline{P}(X_i=x | \mathbf{y}), \overline{P}(X_i=x | \mathbf{y})]$ for every non-evidence variable X_i and every state $x \in D_{X_i}$, where

$$\underline{P}(Z=z | \mathbf{y}) = \min_{\substack{P(X_i | \pi_i) \\ \in \text{ext } K(X_i | \pi_i), \forall i, \pi_i}} \frac{\sum_{\mathbf{x}: x_Z=z, \mathbf{x}_Y=\mathbf{y}} \prod_{i=1}^n P(x_i | \pi_i)}{\sum_{\mathbf{x}: \mathbf{x}_Y=\mathbf{y}} \prod_{i=1}^n P(x_i | \pi_i)}, \quad (2)$$

and $\overline{P}(Z=z | \mathbf{y})$ is defined analogously with $\max$. The optimization is over all combinations of extreme points of the local credal sets — a finite but exponentially large discrete set.

Coordinate-descent reduction. Holding every $P(X_j | \pi_j)$ fixed for $j \neq i$ at a specific extreme point collapses (2) to a fractional linear program in the single remaining distribution $P(X_i | \pi_i)$. Because the feasible region $K(X_i | \pi_i)$ is a polytope, the optimum is attained at one of its (finitely many, pre-enumerated) vertices. Evaluating the objective at each vertex and retaining the best therefore solves the subproblem *exactly and without calling an LP solver*.

Precise-BN subroutine. For any fully fixed assignment $\{P(X_i | \pi_i)\}$, the numerator and denominator of (2) together yield the marginal $P(Z | \mathbf{y})$ of the precise Bayesian network defined by those distributions. Let $\text{VE}(\{P(X_i | \pi_i)\}, Z, \mathbf{y}) \in \Delta^{|D_Z|}$ denote standard (precise) variable elimination applied to that BN, returning the full conditional marginal vector.

Algorithm. Initialize each $P(X_i | \pi_i)$ at the centroid of $\text{ext } K(X_i | \pi_i)$ and compute $p^* = \text{VE}(\cdot)$. A *sweep* visits every pair (X_i, π_i) and, holding all other distributions fixed, replaces $P(X_i | \pi_i)$ with the vertex $v \in \text{ext } K(X_i | \pi_i)$ that most improves $p^*(z)$ in the requested sense (strictly, else keep the incumbent). Sweeps repeat until a full sweep makes no change or the cap T is hit. Running this loop twice (once with sense = min and once with sense = max) for each target state z of each non-evidence variable Z produces a lower/upper pair $(\underline{p}(z), \overline{p}(z))$; because each run also returns a full marginal vector, one pair of sweeps suffices to refine bounds for every state of Z at once via componentwise tracking. The returned intervals are *inner*: since coordinate descent explores only a subset of the feasible vertex combinations, $\underline{P}_{\text{true}} \leq \underline{P}$ and $\overline{P} \leq \overline{P}_{\text{true}}$.

Algorithm 2 ApproxLP — Compact Form

Require: Credal factors $\{\text{ext } K(X_i \mid \pi_i)\}$, evidence $\mathbf{Y} = \mathbf{y}$, iteration cap T

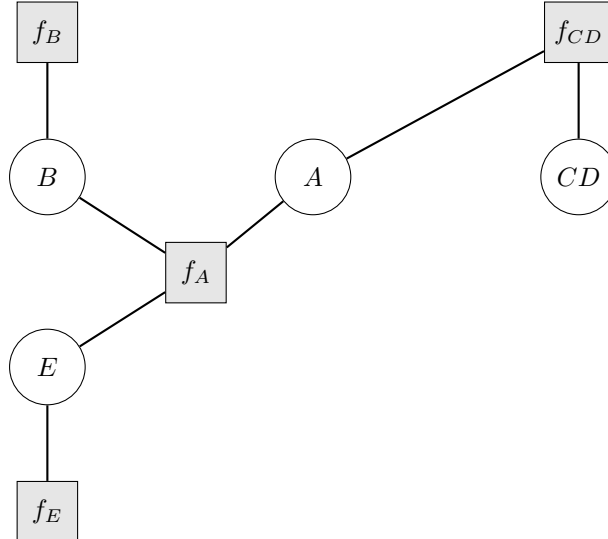
Ensure: Inner bounds $[\underline{P}(X_i=x \mid \mathbf{y}), \bar{P}(X_i=x \mid \mathbf{y})]$ for every $X_i \notin \mathbf{Y}$, $x \in D_{X_i}$

```
1: for each non-evidence variable  $Z$  do
2:    $\underline{p}, \bar{p} \leftarrow \mathbf{1}, \mathbf{0}$   $\triangleright$  componentwise over  $D_Z$ 
3:   for  $\text{sense} \in \{\min, \max\}$  do
4:      $P(X_i \mid \pi_i) \leftarrow \text{centroid}(\text{ext } K(X_i \mid \pi_i))$  for all  $(X_i, \pi_i)$ ;  $p^* \leftarrow \text{VE}(\{P\}, Z, \mathbf{y})$ 
5:     for sweep  $t = 1, \dots, T$  do
6:       for each  $(X_i, \pi_i)$  do
7:          $v^* \leftarrow \arg \text{sense}_{v \in \text{ext } K(X_i \mid \pi_i)} \text{VE}(\{P\}[P(X_i \mid \pi_i) \leftarrow v], Z, \mathbf{y})(z)$ 
 $\triangleright z = \text{target state this run is tracking}$ 
8:         if  $v^*$  strictly improves  $p^*(z)$  then  $P(X_i \mid \pi_i) \leftarrow v^*$ ,  $p^* \leftarrow \text{VE}(\{P\}, Z, \mathbf{y})$ 
9:       end for
10:      if sweep made no change then break
11:    end for
12:     $\underline{p} \leftarrow \min(\underline{p}, p^*)$ ,  $\bar{p} \leftarrow \max(\bar{p}, p^*)$ 
13:  end for
14:  Record  $[\underline{p}(z), \bar{p}(z)]$  for every  $z \in D_Z$ 
15: end for
```

4 Detailed Example: Alarm Network

We trace ApproxLP on the alarm network (`alarm.lcn`) with variables B, E, A, CD and the DAG $B \rightarrow A, E \rightarrow A, A \rightarrow CD$.

4.1 Factor Graph



4.2 Credal Factors (Extreme Points)

f_B : $K(B)$. $v_1 = [0.80, 0.20]$, $v_2 = [0.90, 0.10]$.

f_E : $K(E)$. $v_1 = [0.90, 0.10]$, $v_2 = [0.95, 0.05]$.

f_A : $K(A | B, E)$. 4 parent configs, 2 vertices each:

Config	Vtx	$P(A=0)$	$P(A=1)$
$B=0, E=0$	v_1	0.9556	0.0444
$B=0, E=0$	v_2	1.0000	0.0000
$B=1, E=0$	v_1	0.0000	1.0000
$B=1, E=0$	v_2	1.0000	0.0000
$B=0, E=1$	v_1	0.0000	1.0000
$B=0, E=1$	v_2	1.0000	0.0000
$B=1, E=1$	v_1	0.0000	1.0000
$B=1, E=1$	v_2	1.0000	0.0000

f_{CD} : $K(CD | A)$. $A=0$: 4 vertices; $A=1$: 11 vertices (see companion documents for full listing).

4.3 Query 1: $P(B=1)$, No Evidence

Step 1: Initialization. Each distribution is set to the center (average) of its credal set:

$$\begin{aligned}
 P(B) &= \frac{1}{2}([0.80, 0.20] + [0.90, 0.10]) = [0.85, 0.15], \\
 P(E) &= \frac{1}{2}([0.90, 0.10] + [0.95, 0.05]) = [0.925, 0.075], \\
 P(A | B=0, E=0) &= \frac{1}{2}([0.9556, 0.0444] + [1.0, 0.0]) = [0.9778, 0.0222].
 \end{aligned}$$

(Other parent configs for A are similarly averaged.)

Step 2: Initial evaluation. Run VE with these fixed distributions:

$$P(B=1) = 0.1500 \quad (\text{the center value for } B).$$

Step 3: Coordinate descent (sense = min). Iteration 0:

Try variable B , config ():

- Set $P(B) = v_1 = [0.80, 0.20]$, run VE: $P(B=1) = 0.2000$. Worse than 0.1500 $\rightarrow$ reject.
- Set $P(B) = v_2 = [0.90, 0.10]$, run VE: $P(B=1) = 0.1000$. Better $\rightarrow$ **accept**. Update incumbent: $p^* = 0.1000$.

Try variable E , config ():

- $v_1 = [0.90, 0.10]$: VE gives $P(B=1) = 0.1000$. No improvement.
- $v_2 = [0.95, 0.05]$: VE gives $P(B=1) = 0.1000$. No improvement.

Since B is a root node, $P(B)$ directly determines $P(B=1)$ regardless of other distributions. The algorithm correctly identifies $v_2 = [0.90, 0.10]$ as the minimizer in one iteration.

Other variables (A, CD) have no effect on $P(B=1)$ after B 's vertex is fixed.

Iteration 1: No improvement found $\rightarrow$ converged.

Result (min): $\underline{P}(B=1) = 0.1000$.

Coordinate descent (sense = max). Analogous: selecting $v_1 = [0.80, 0.20]$ gives $\bar{P}(B=1) = 0.2000$. Converges in 1 iteration.

Final result:

$$P(B=1) \in [0.1000, 0.2000].$$

4.4 Query 2: $P(A=1 \mid B=0, E=0)$

Initialization. With evidence $B=0, E=0$, the only relevant parent config for A is $(B=0, E=0)$. The initial center distribution is:

$$P(A \mid B=0, E=0) = [0.9778, 0.0222].$$

Initial evaluation. VE with fixed distributions and evidence gives $P(A=1 \mid B=0, E=0) = 0.0222$.

Coordinate descent (sense = min). Try A , config $(B=0, E=0)$:

- $v_1 = [0.9556, 0.0444]$: VE gives $P(A=1) = 0.0444$. Worse $\rightarrow$ reject.
- $v_2 = [1.0000, 0.0000]$: VE gives $P(A=1) = 0.0000$. Better $\rightarrow$ **accept**.

Converged: $\underline{P}(A=1 \mid B=0, E=0) = 0.0000$.

Coordinate descent (sense = max). Try A , config $(B=0, E=0)$:

- $v_1 = [0.9556, 0.0444]$: $P(A=1) = 0.0444$. Better $\rightarrow$ **accept**.
- $v_2 = [1.0, 0.0]$: $P(A=1) = 0.0000$. Worse $\rightarrow$ reject.

Converged: $\bar{P}(A=1 \mid B=0, E=0) = 0.0444$.

Final result:

$$P(A=1 \mid B=0, E=0) \in [0.0000, 0.0444].$$

Both results match exact VE, confirming that ApproxLP finds the global optimum on this network.

5 Properties

Inner bounds. ApproxLP produces *inner* bounds: the returned interval $[\underline{P}, \bar{P}]$ is guaranteed to satisfy $\underline{P}_{\text{true}} \leq \underline{P}$ and $\bar{P} \leq \bar{P}_{\text{true}}$. This is because coordinate descent explores a subset of the feasible extreme point combinations.

Monotonic convergence. Each iteration either improves the objective or stops. The objective is bounded, so convergence is guaranteed in a finite number of steps.

Local optima. Coordinate descent may get stuck at a local optimum: the current combination of vertices may be locally optimal (no single vertex swap improves the objective) but globally suboptimal (a simultaneous swap of multiple vertices would improve it). In practice, this is rarely an issue for small to medium networks.

Speed. Each VE evaluation operates on a *precise* Bayesian network (single distributions, no sets of functions), which is orders of magnitude faster than credal VE. For the alarm network: ApproxLP takes 0.007 s vs. credal VE’s 3.76 s—a $500\times$ speedup.

6 Comparison with Other Methods

Property	Exact VE	ε -VE	CTE	IBP	ApproxLP
Exact	Yes	No	Yes	No	No
Bound type	Tight	Outer	Tight	Outer	Inner
All marginals	No	No	Yes	No	No
Multi-valued	Yes	Yes	Yes	Yes	Yes
Speed (alarm)	3.76 s	0.009 s	50 s	0.01 s	0.007 s

ApproxLP and IBP produce complementary bounds: ApproxLP’s inner bounds sit inside the true interval, while IBP’s outer bounds contain it. Together they can bracket the true interval from both sides.

7 Complexity

Each coordinate descent iteration visits all variables, all parent configs, and all extreme points. For each vertex tried, a full VE is run on a precise BN.

- **Per VE evaluation:** $O(n \cdot k^{w^*})$ where n is the number of variables, k the max domain size, and w^* the treewidth.
- **Per iteration:** $O(\sum_i \sum_{\pi} |\text{ext } K(X_i \mid \pi)|) \times O(n \cdot k^{w^*})$ VE evaluations.
- **Total:** $O(T \cdot V_{\text{total}} \cdot n \cdot k^{w^*})$ where T is the number of iterations and V_{total} is the total number of extreme points across all credal sets.

In practice, T is typically 2–5 (fast convergence), making ApproxLP very efficient.

8 References

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